SE Practices 1 | SEP1

Summary

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WARN:
This document is still under development and will be completed around 1.2.2027

1. OBJECT-ORIENTED ANALYSIS AND DESIGN

“Responsibility-driven design”

1) Requirements analysis: What should the product do?
   – Software Requirements Specification (SRS)
2) Domain analysis: What is the problem domain?
   – Static View: Domain Model (UML notation: Class Diagram)
   – Dynamic View: Behavioural Models (UML notation: Interaction Diagram, State Machine Diagram, Activity Diagram, etc. depending on the domain)

TODO:
diagrams oo slides 6

TODO:
oo slides in general

Analysis: Investigation of the objects in the problem domain

Design: Defining software objects and how they collaborate to fulfill the requirements

1) Define Use Cases
2) Define a Domain Model
3) Assign Object Responsibilities and Draw Interaction Diagrams
4) Define Design Class Diagrams

2. UNIFIED MODELING LANGUAGE (UML)

Visual language for specifying, constructing and documenting the artifacts of systems.

Perspectives to apply UML (from abstract to specific): Conceptual, Specification, Implementation

UML Notation for Domain Models:
– Classes for sets of similar objects in the problem domain
– Attributes for properties of those objects
– Associations for relationships between those objects
– Generalisations in between classes with a “each X is a Y” relationship

3. UNIFIED PROCESS (UP)

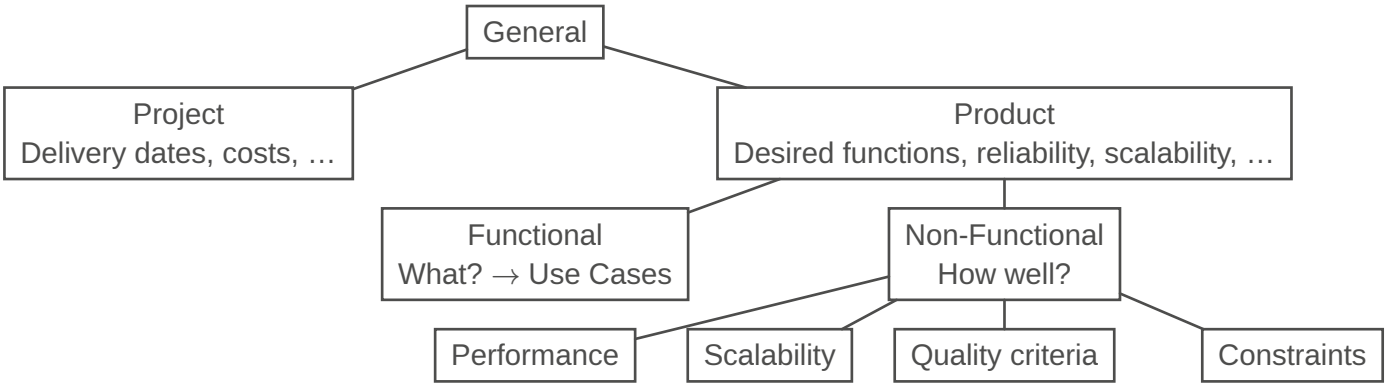
An iterative software development process for building object-oriented systems.

Software development process: describes an approach to building, deploying and possibly maintaining software.

4. ITERATIVE DEVELOPMENT

5. DOMAIN MODELS

6. REQUIREMENTS



Discipline: a systematic approach to finding, documenting, organizing, and tracking the changing requirements of a system

Work Products: Software Requirement Specification (SRS) = Capabilities/Conditions that the software must fulfil

6.1. USE CASES

"A set of use-case instances, where each instance is a sequence of actions a system performs that yields an observable result of value to a particular actor."

– A textual process description with a concrete purpose
– Documents a story of some actor using a system to meet goals
– Places requirements within the context of a user with goals.
– Simple & familiar for the average stakeholder



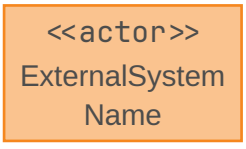
A use case is typically composed of:
– A main success scenario, and
– Multiple extensions / alternative scenarios

6.1.1. Actors

An actor is something with behaviour, such as a person (identified by a role), computer system, or organization.

Actors are associated to roles, and not to physical entities





Primary Actors: Actors whose goals are fulfilled through using services of the SuD.
E.g., cashier in "Process Sale"

Supporting Actors: Provides a service to the SuD
E.g., Credit card billing system in "Process Sale"

Offstage Actors: Has an interest in the behaviour of the use case, but is not primary or supporting.
E.g., Government tax agency in "Process Sale"

6.1.2. Formats

Brief: Short, one-paragraph summary, usually of the main success scenario

Casual: Multiple paragraphs that cover various scenarios

Fully dressed: All scenarios, steps and variations written in detail, in a specific form, with supporting sections, such as preconditions and success guarantees.

TODO:
W3 S12

6.1.3. Elementary Business Processes (EBP)

Use Cases should be identified at the EBP level:

A task performed by one person in one place at one time, (single session) in response to a business event, which adds measurable business value (Boss Test) and leaves the data in a consistent state

6.1.4. Finding Use Cases

1) Fix the system boundary: Software, HW/SW-System, Entire Organisation?
2) Identify primary actors and their goals: At the EBP level
3) Write down the use cases – first in the brief Format

6.1.5. System boundary

TODO:
W3 S19+